UCS3461 Foundations of Artificial Intelligence

BAYES NETWORKS (BELIEF NETWORKS)

Semester 4, B.E. Computer Science and Engineering
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Bayes Nets

1. Representation
2. Exact inference
   - Enumeration (always exponential complexity)
   - Variable Elimination (worst-case exponential complexity, often better)
   - Inference is NP-hard in general
3. Approximate Inference

Model and Inference

Probability Model

- A probability model is a joint distribution on a set of random variables, $P(X_1, \ldots, X_n)$
- Written as a table, this would be exponential in n
- If the variables are independent, joint distribution is the product of marginal distributions
  - $P(x, y) = P(x)P(y)$ or $P(x|y) = P(x)$

  Eg: probability model for n coins represented by n numbers instead of 2^n
- Independence is rare in practice: within a domain, most variables correlated
- Conditional independence is much more common.
  - Traffic and Umbrella are conditionally independent given Rain
  - Alarm and Fire are conditionally independent given Smoke

Bayes Net: Idea

- Bayes Net (Belief Network): a technique for describing complex joint distributions (models) using simple local distributions (conditional distributions)
- A subset of the general class of graphical models
- Takes advantage of local causality:
  - the world is composed of many variables,
  - each variable interacts locally with a few others

Graphical Models

- Nodes: random variables (with domains)
  - Values assigned (observed) or
  - Values unassigned (unobserved)
- Edges: interactions
  - Indicate "direct influence" between variables
  - Formally: encode conditional independence
- Imagine that arrows mean direct causation
  - In general, they don't mean causation!
  - Indicate only correlation!
- Example: Coin Flips
  - N independent coin flips
  - No interactions between variables: absolute independence

Examples

- Traffic: variables
  - T: There is traffic
  - U: I'm holding my umbrella
  - R: It rains

$$\begin{tikzcd} & R \arrow[dl] \arrow[dr] & \\ T & & U \end{tikzcd}$$

- Smoke alarm: variables
  - F: There is fire
  - S: There is smoke
  - A: Alarm sounds

$$F \longrightarrow S \longrightarrow A$$

Bayes Net

Bayes Net Syntax

- A Directed Acyclic Graph (DAG)
- A set of nodes: one node per random variable X_i
- A conditional distribution for each RV, given its parent variables in the graph
  - CPT (Conditional Probability Table): each row is a distribution for child given a configuration of its parents
- A Bayes Net = Topology (graph) + Local conditional distributions

Building Bayes Net: Example

Variables

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!

Building Bayes Net: Alarm Network

P(B =1)
0.001

Burglary

Earthquake

P(E=1)
0.002

Alarm

B	E	P(A=1|B,E)
0	0	0.001
0	1	0.29
1	0	0.94
1	1	0.95

JohnCalls

A	P(J =1|A)
0	0.05
1	0.9

MaryCalls

A	P(M = 1|A)
0	0.01
1	0.7

General Formula for Sparse BNs

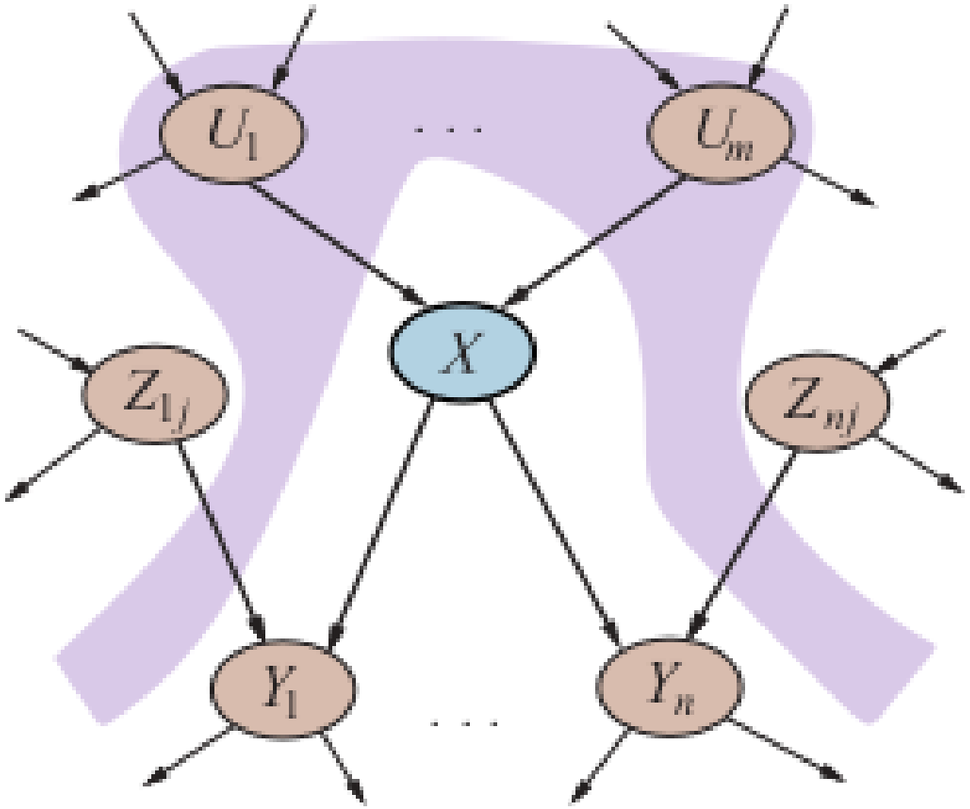
- Suppose
  - n random variables
  - Maximum domain size is d
  - Maximum number of parents is k
- Full joint distribution has size $O(d^n)$
- Bayes net has size $O(nd^k)$

$$O(nd^k) \ll O(d^n)$$

- Linear scaling with n as long as causal structure is local

Conditional Independence Semantics

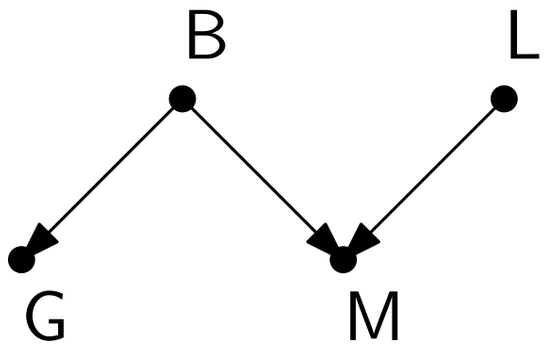
- Every variable is conditionally independent of its non-descendants (topological predecessors) given its parents
- Conditional independence semantics $\equiv$ global semantics

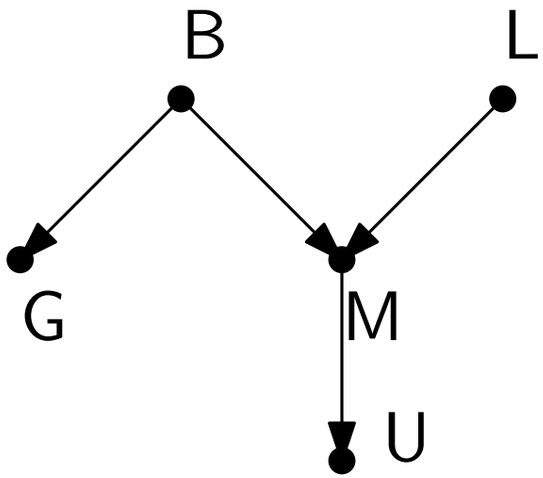


$$P(X_i|\text{Non-descendants}(X_i), \text{Parents}(X_i)) = P(X_i|\text{Parents}(X_i))$$

Conditional Independence Semantics

$P(b) = 0.95 \qquad P(l) = 0.7$





B	P(g|B)
0	0.1
1	0.95

B	E	P(m|B,L)
0	0	0.0
0	1	0.0
1	0	0.05
1	1	0.9

$$P(m|G, B, L, U) = P(m|B, L, U)$$

$$P(G, M, B, L) = P(M|G, B, L)P(G|B, L)P(B|L)P(L)$$

$$P(M|G, B, L) = P(M|B, L)$$

$$P(G|B, L) = P(G|B)$$

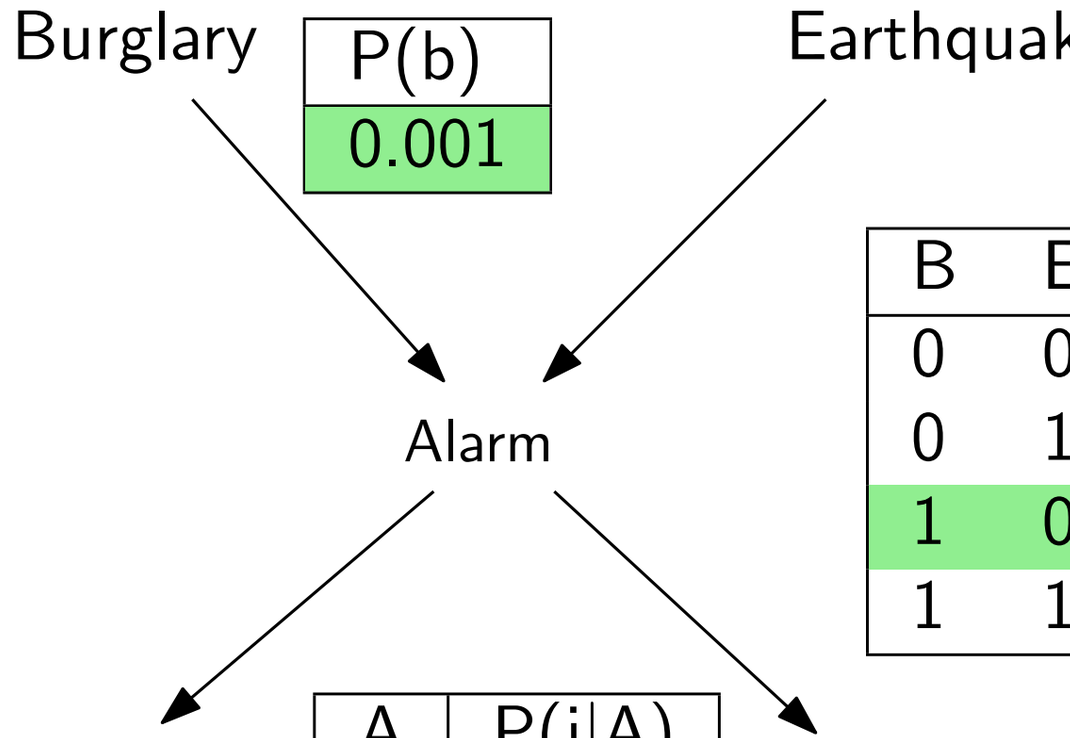
$$P(G, M, B, L) = P(M|B, L)P(G|B)P(B)P(L)$$

Bayes Net Semantics

- Bayes nets encode joint distributions as a product of conditional distributions on each variable:
- To construct a Bayesian network for a given set of variables, draw edges from cause variables to immediate effects. In almost all cases, doing so results in a Bayesian network (whose conditional-independence implications are accurate).

$$P(X_1, \ldots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$$

Example: Alarm Network



P(b)
0.001

P(e)
0.002

B	E	P(a|B,E)
0	0	0.001
0	1	0.29
1	0	0.94
1	1	0.95

A	P(j|A)
0	0.05
1	0.9

A	P(m|A)
0	0.01
1	0.7

$$P(B=1,E=0,A=1,J=0,M=0) = P(b,\neg e,a,\neg j,\neg m)$$
$$= P(b)P(\neg e)P(a|b,\neg e)P(\neg j|a)P(\neg m|a)$$
$$= .001 \times .998 \times .94 \times .1 \times .3 = .000028$$

Probabilities in BNs

- Chain rule (valid for all distributions):
  $P(X_1, \ldots, X_n) = \prod_i P(X_i | X_1, \ldots, X_{i-1})$
- Topologically order the variables $X_1, \ldots, X_n$ – each variable X_i depends only on its predecessors $X_1, \ldots, X_{i-1}$.
- Conditional independences hold:
  $P(X_i | X_1, \ldots, X_{i-1}) = P(X_i | \text{Parents}(X_i))$
- Consequence: $P(X_1, \ldots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$
- We guaranteed that

$$P(X_1, \ldots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$$

  results in a proper joint distribution.

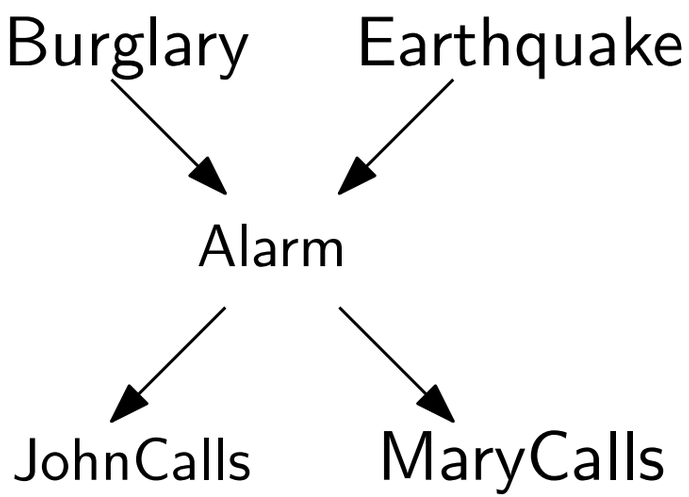
Causality?

- When Bayes Nets reflect the true causal patterns,
  - Often simpler (fewer parents, fewer parameters)
  - Often easier to assess probabilities
- What do the arrows really mean?
  - Topology may happen to encode causal structure
  - Topology really encodes conditional independence:

$$P(X_i|X_1, \ldots, X_{i-1}) = P(X_i|\text{Parents}(X_i))$$

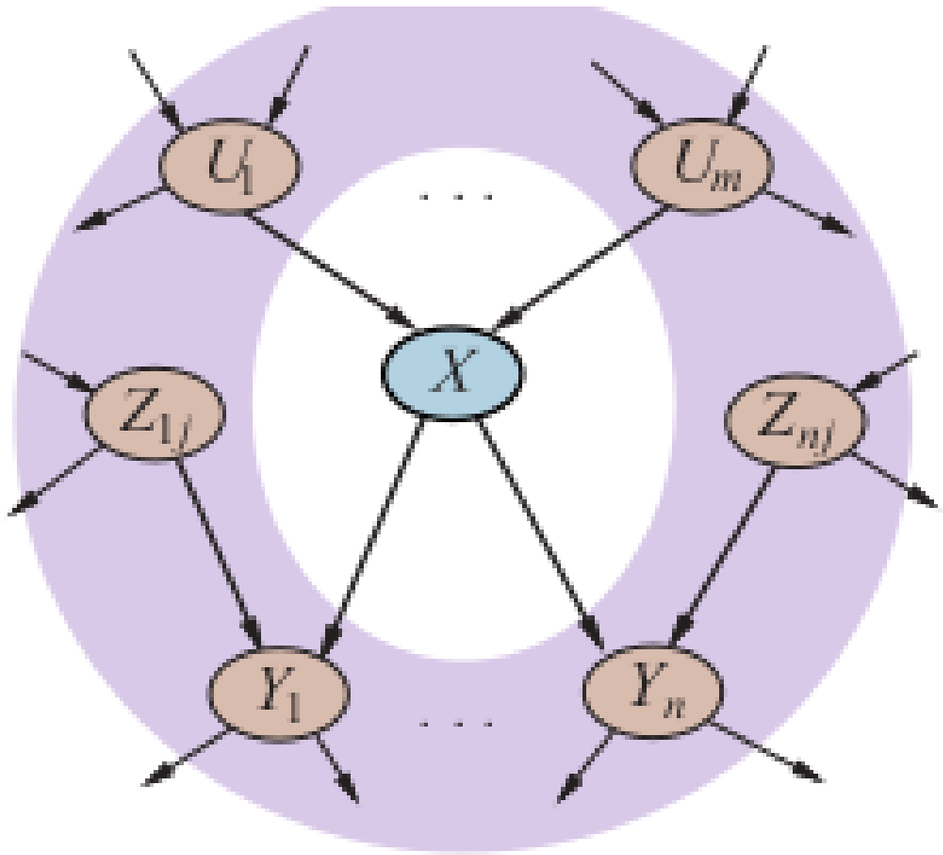
Example

- JohnCalls independent of Burglary given Alarm? Yes
- JohnCalls independent of MaryCalls given Alarm? Yes
- Burglary independent of Earthquake? Yes
- Burglary independent of Earthquake given Alarm? No!
  - Given that the alarm has sounded, both burglary and earthquake become more likely
  - But if we then learn that a burglary has happened, the alarm is explained away and the probability of earthquake drops back



Markov Blanket

- A variable's Markov blanket consists of parents, children, children's other parents
- Every variable is conditionally independent of all other variables given its Markov blanket



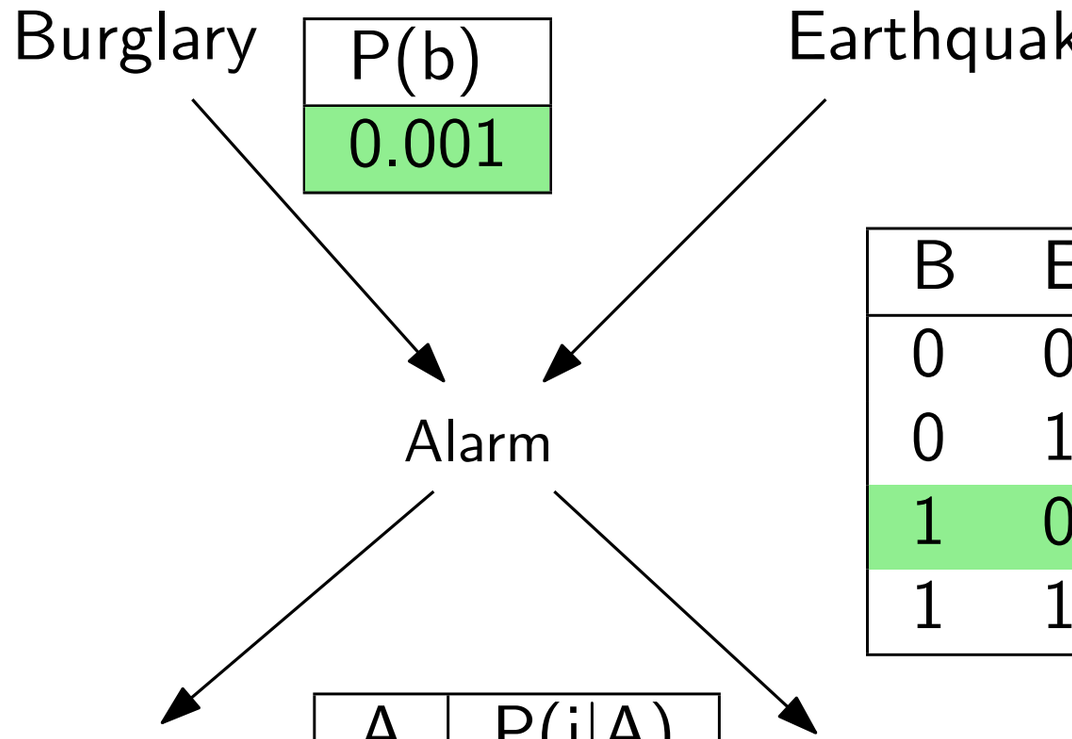
Exact Inference: Enumeration

Bayes Net Semantics

Bayes Nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1, \ldots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$$

Example: Alarm Network



P(b)
0.001

P(e)
0.002

B	E	P(a|B,E)
0	0	0.001
0	1	0.29
1	0	0.94
1	1	0.95

A	P(j|A)
0	0.05
1	0.9

A	P(m|A)
0	0.01
1	0.7

$$P(B=1,E=0,A=1,J=0,M=0) = P(b,\neg e,a,\neg j,\neg m)$$
$$= P(b)P(\neg e)P(a|b,\neg e)P(\neg j|a)P(\neg m|a)$$
$$= .001 \times .998 \times .94 \times .1 \times .3 = .000028$$

Inference

- Calculating some useful quantity from a probability model (joint probability distribution)
  - Posterior marginal probability: $P(Q|x_1, \ldots, x_k)$.
    - E.g. what disease might I have?
  - Most likely explanation: $\text{argmax}_{q,r,s} P(Q = q, R = r, S = s|x_1, \ldots, x_k)$.
    - E.g., what did he say?
- Query: $P(\mathbf{Q}|\mathbf{X}, \mathbf{Y})$
  - Query variables $\mathbf{Q}$,
  - Observed variables (evidence) $\mathbf{X}$,
  - Unobserved variables (hidden variables) $\mathbf{Y}$

Exact Inference

- Enumeration (always exponential complexity)
- Variable elimination (worst-case exponential complexity, often better)
- Inference is NP-hard in general

Inference by Enumeration

- A marginal distribution (maybe conditional) computed by summing entries from the joint distribution
  - $P(Q|e) = \alpha P(Q, e) = \alpha \sum_H P(Q, e, H)$
- Entries for the joint distribution can be obtained from a BN by multiplying the corresponding conditional probabilities

$$
\begin{aligned}
P(B|j,m) &= \alpha P(B,j,m) \\
&= \alpha \sum_{E,A} P(B,E,A,j,m) \\
&= \alpha \sum_{E,A} P(B)P(E)P(A|B,E)P(j|A)P(m|A)
\end{aligned}
$$

- Inference in Bayes Nets is computing sums of products of numbers!
- Problem: sums of exponentially many products $O(n2^n)$!

Enumeration

Algorithm: Enumerate(vars, assign, factors)

Input:

vars: Set of unassigned variables

assign: Map of assigned variables with their values

factors: Set of factors

Output: $\sum_{\text{var} \in \text{vars}} \prod_{f \in \text{factors}} f[\text{assign}]$

if factors $= \{\}$ **then**

  return 1

else if there is a f $\in$ factors that can be evaluated with assign **then**

  return f[assign] $\times$ Enumerate (vars, assign, factors $-\{f\}$)

else

  var $\leftarrow$ select a variable in vars

  return $\displaystyle\sum_{\text{val} \in \text{domain(var)}}$ Enumerate(vars $-\{\text{var}\}$, assign $\cup\{\text{var} = \text{val}\}$, factors)

Enumeration

Algorithm: Enumerate(vars, assign)

Input:

vars: Set of unassigned variables

assign: Map of (var = val) – var is an assigned variable, val its value

Output: $\sum_{\text{var} \in \text{vars}} \prod_{f \in \text{factors}} f[\text{assign}]$

if vars $= \{\}$ **then**

  return 1

var $\leftarrow$ vars[0]

if var is in assign **then**

  return P(assign[var]|parents(var)) $\times$ Enumerate (vars, assign)

else

  return $\displaystyle\sum_{\text{val} \in \text{domain(var)}}$ Enumerate(vars[1:], assign $\cup\{\text{var} = \text{val}\}$)

Exact Inference: Variable Elimination

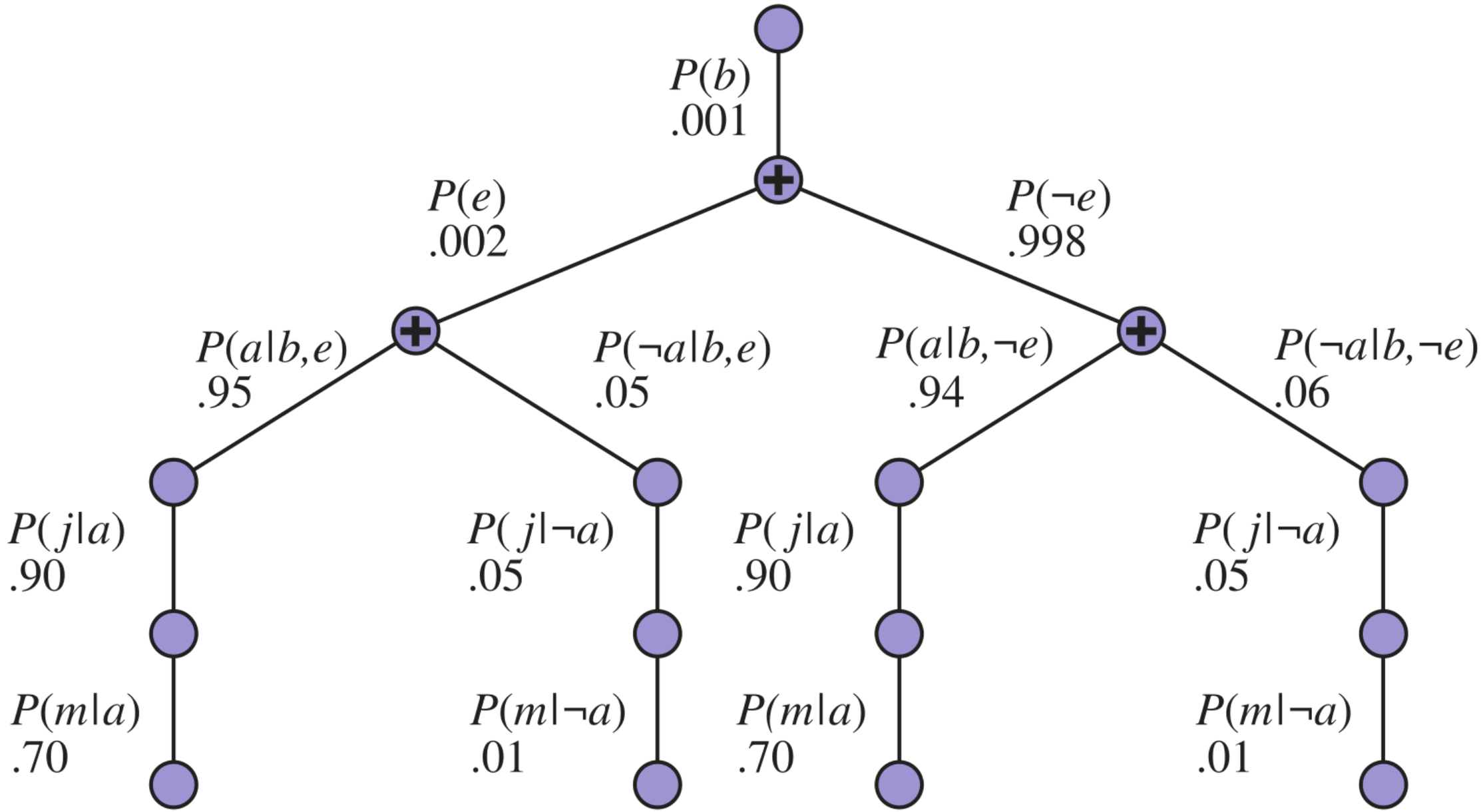
Factorize (Repeated Calculation)

- Consider $uwy + uwz + uxy + uxz + vwy + vwz + vxy + vxz$
  - Terms: 8, add: 8, multiply: 16
  - Lots of repeated subexpressions!
- Rewrite as $(u + v)(w + x)(y + z)$
  - Add: 3, multiply: 2

$$\sum_{E,A} P(b)P(E)P(A|b,E)P(j|A)P(m|A)$$

$$\begin{aligned}
=&P(b)P(e)P(a|b,e)P(j|a)P(m|a)+\\
&P(b)P(\neg e)P(a|b,\neg e)P(j|a)P(m|a)+\\
&P(b)P(e)P(\neg a|b,e)P(j|\neg a)P(m|\neg a)+\\
&P(b)P(\neg e)P(\neg a|b,\neg e)P(j|\neg a)P(m|\neg a)
\end{aligned}$$

- Lots of repeated subexpressions!


P(b)
.001
P(e)
.002
P(¬e)
.998
P(a|b,e)
.95
P(¬a|b,e)
.05
P(a|b,¬e)
.94
P(¬a|b,¬e)
.06
P(j|a)
.90
P(j|¬a)
.05
P(j|a)
.90
P(j|¬a)
.05
P(m|a)
.70
P(m|¬a)
.01
P(m|a)
.70
P(m|¬a)
.01

Variable Elimination

- Move summations inward as far as possible

$$P(b|j,m) = \alpha \sum_{E,A} P(b)P(E)P(A|b,E)P(j|A)P(m|A)$$

$$= \alpha P(b) \sum_{E} P(E) \sum_{A} P(A|b,E)P(j|A)P(m|A)$$

- Do the calculation from the inside out
  - Sum over A first, then sum over E
  - Problem: $P(A|b,E)$ is not a single number, it is an array of numbers depending on the values of A and E – factors

Operation 1: Pointwise Product

- First basic operation: pointwise product of factors (not matrix multiply!)
  - New factor has union of variables of the two original factors
  - Each entry is the product of the corresponding entries from the original factors
- Example: $P(J|A) \times P(A) = P(A, J)$

A	$P(A)$
1	0.1
0	0.9

$\times$

J	A	$P(J\|A)$
1	1	0.9
1	0	0.05
0	1	0.1
0	0	0.95

$=$

J	A	$P(J\|A)$
1	1	0.09
1	0	0.045
0	1	0.01
0	0	0.855

Pointwise Product: Example

$f(X,Y) \times g(Y,Z) = h(X,Y,Z)$

X	Y	$f(X,Y)$	Y	Z	$g(X,Y)$	X	Y	Z	$h(X,Y,Z)$
1	1	.3	1	1	.2	1	1	1	$.3 \times .2 = .06$
1	0	.7	1	0	.8	1	1	0	$.3 \times .8 = .24$
0	1	.9	0	1	.6	1	0	1	$.7 \times .6 = .42$
0	0	.1	0	0	.4	1	0	0	$.7 \times .4 = .28$
						0	1	1	$.9 \times .2 = .18$
						0	1	0	$.9 \times .8 = .72$
						0	0	1	$.1 \times .6 = .06$
						0	0	0	$.1 \times .4 = .04$

Making Larger Factors

- Example: $P(U, V) \times P(V, W) \times P(W, X) = P(U, V, W, X)$
- Sizes: $[10, 10] \times [10, 10] \times [10, 10] = [10, 10, 10, 10]$
- 300 numbers blow up to 10,000 numbers!
- Factor blowup can make VE very expensive

Operation 2: Sum out a Variable

- Second basic operation: sum out (eliminate) a variable from a factor
  - Shrinks a factor to a smaller one
- Example: $\sum_J P(A, J) = P(A, j) + P(A, \neg j) = P(A)$

J	A	$P(A, J)$
1	1	0.09
1	0	0.045
0	1	0.01
0	0	0.855

$\xrightarrow{\text{Sum out } J}$

$P(A)$	
1	0.1
0	0.9

Operation 2: Sum out from a Product of Factors

- Project the factors each way first, then sum the products
- Example:

$$\sum_A P(A|b,E) \times P(j|A) \times P(m|A)$$
$$= P(a|b,E) \times P(j|a) \times P(m|a) + P(\neg a|b,e) \times P(j|\neg a) \times P(m|\neg a)$$

Variable Elimination

- Query: $P(Q|E_1 = e_1, \ldots, E_k = e_k)$
- Start with initial factors:
  - Local CPTs (instantiated by evidence)
- While there are still hidden variables (neither Q nor evidence):
  - Pick a hidden variable H
  - Join all factors mentioning H
  - Eliminate (sum out) H
- Join all remaining factors and normalize

Example

$$
\begin{aligned}
& P(B, E, A, j, m) \\
& \propto P(B) \quad P(E) \quad P(A|B,E) \quad P(j|A) \quad P(m|A) \\
& = f_1(B) \quad f_2(E) \quad f_3(A,B,E) \quad f_4(A) \quad f_5(A) \\
P(B|j,m) & = f_1(B) \quad \sum_E f_2(E) \quad \sum_A f_3(A,B,E) \quad f_4(A) \quad f_5(A)
\end{aligned}
$$

$$\langle \text{Choose } A : f_3(A,B,E) \quad f_4(A) \quad f_5(A) \xrightarrow{\times, \sum} f_6(B,E) \rangle$$

$$= f_1(B) \quad \sum_E f_2(E) \quad f_5(B,E)$$

$$\langle \text{Choose } E : f_2(E) \quad f_5(B,E) \xrightarrow{\times, \sum} f_7(B) \rangle$$

$$= f_1(B) \quad f_7(B)$$

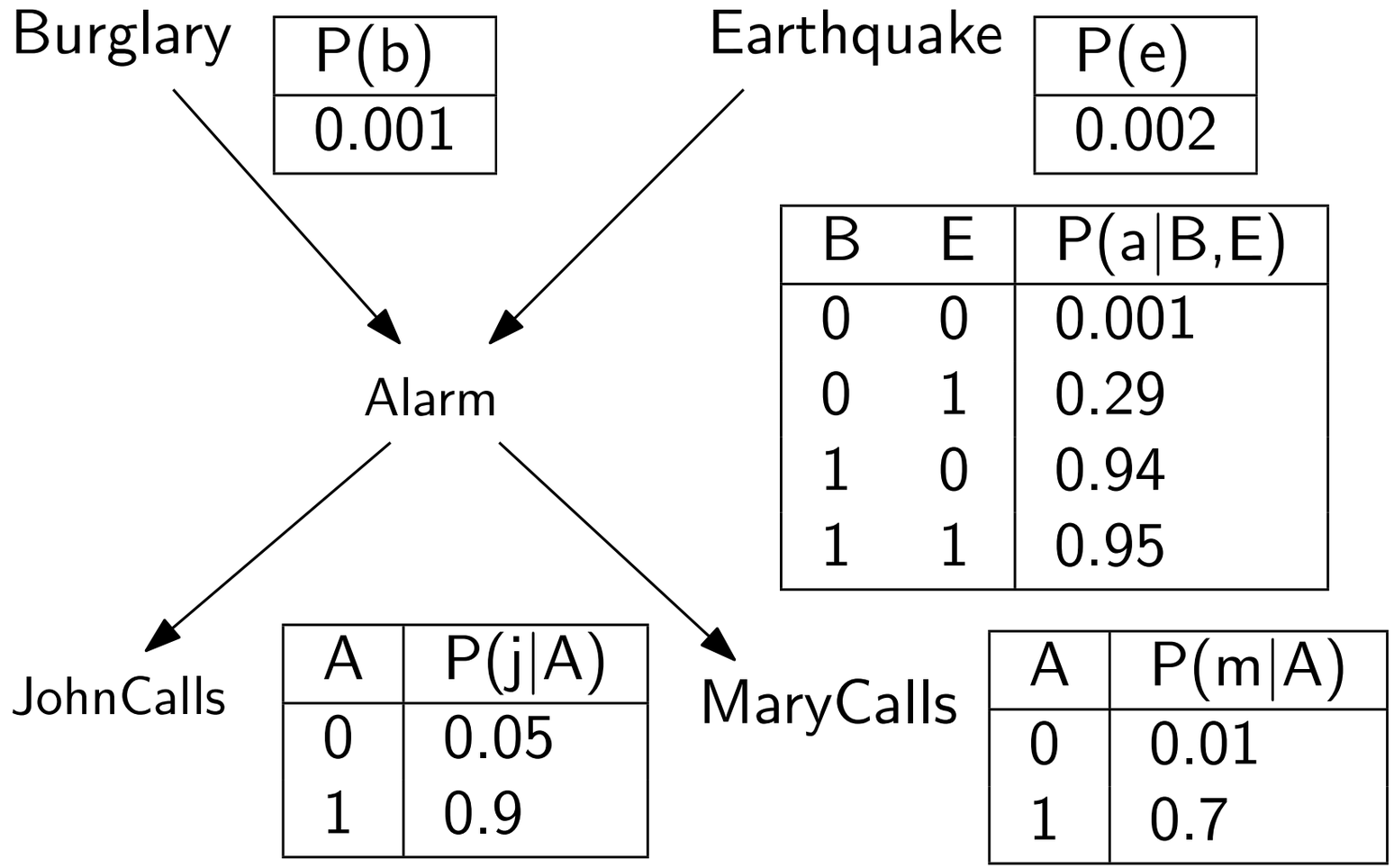
$$\langle \text{Choose } B : f_1(B) \quad f_7(B) \xrightarrow{\times} P(b,j,m) \rangle$$

$$= c$$

Example

- Query $P(b|j,m)$
  $P(b) \quad P(E) \quad P(A|b,E) \quad P(j|A) \quad P(m|A)$
- Choose A
  - $P(A|b,E) \quad P(j|A) \quad P(m|A) \xrightarrow{\times,\sum} P(j,m|b,E)$
  - $P(b) \quad P(E) \quad P(j,m|b,E)$
- Choose E
  - $P(E) \quad P(j,m|b,E) \xrightarrow{\times,\sum} P(j,m|b)$
  - $P(b) \quad P(j,m|b)$
- Finish with B
  - $P(b) \quad P(j,m|b) \xrightarrow{\times} P(j,m,b)$

Example: Alarm Network



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A	P(j|A)
0	0.05
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A	P(m|A)
0	0.01
1	0.7

Variable Elimination

Algorithm: VariableElimination_BN(vars, factors, evidence, Q)

Input: vars: Set of all variables, factors: Set of all CPDs, evidence: Map of (var = val) , Q: Query variable

Output: Posterior distribution of Q

foreach X $\in$ vars $-\{Q\}$ in some elimination ordering **do**

factors' $\leftarrow$ {f $\in$ factors | f involves X }

if X is in evidence **then**

assign evidence[X] to X in each f $\in$ factors

else

factors $\leftarrow$ factors $-$ factors' $\cup\{\sum_{\text{X}} \prod_{f \in \text{ factors'}} f\}$

T $\leftarrow \prod_{f \in \text{ factors}} f$

return $\dfrac{T}{\sum_Q T}$

Variable Elimination

Algorithm: VariableElimination_BN(Vs, Ps, Es, Q)

Input: Vs: Set of all variables, Ps: Set of all CPs, Es: Observations, Q: Query variable

Output: Posterior distribution of Q

Fs := Ps
foreach X $\in$ Vs $\setminus$ Q in some elimination ordering **do**
    Rs := $\{F \in Fs \mid F$ involves $X\}$
    if X $\in$ Es **then**
        assign observed value in e to X in each F $\in$ Rs
    else
        let T be the product of the factors in Rs
        construct an explicit representation of $N = \sum_Z T$
        Fs := Fs $\setminus$ Rs $\cup\{N\}$
let T be the product of the factors in Fs
N := $\sum_Y T$
return T/N

Order Matters

- Order the terms Z, A, B, C, D
  - $P(D) = \alpha \sum_{z,a,b,c} P(z)P(a|z)P(b|z)P(c|z)P(D|z)$
  - $= \alpha \sum_z P(z) \sum_a P(a|z) \sum_b P(b|z) \sum_c P(c|z)P(D|z)$
  - Largest factor has 2 variables (D, Z)
- Order the terms A, B, C, D, Z
  - $P(D) = \alpha \sum_{a,b,c,z} P(a|z)P(b|z)P(c|z)P(D|z)P(z)$
  - $= \alpha \sum_a \sum_b \sum_c \sum_z P(a|z)P(b|z)P(c|z)P(D|z)P(z)$
  - Largest factor has 4 variables (A, B, C, D)
- In general, with n leaves, factor of size 2^n

VE: Time and Space Complexity

- Determined by the largest factor (and it's space that kills you)
- Elimination ordering can greatly affect the size of the largest factor.
- Does there always exist an ordering that only results in small factors? No

Summary

- Bayes Net (BN) encodes a joint distribution
- How to answer queries, i.e., compute conditional probabilities of queries given evidence

Inference

$$
\begin{aligned}
&P(S=1|A=1)\\
&= \frac{P(S=1, A=1)}{P(A=1)}\\
&= \frac{\sum_{B,R} P(A=1, B, R, S=1)}{\sum_{B,R,S} P(A=1, B, R, S)}\\
&= \frac{\sum_{B,R} P(A=1|R, S=1)P(B|R)P(R)P(S=1)}{\sum_{B,R,S} P(A=1|R, S)P(B|R)P(R)P(S)}\\
&= \frac{\sum_{R} P(A=1|R, S=1)P(R)P(S=1)}{\sum_{R,S} P(A=1|R, S)P(R)P(S)}\\
&= \frac{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1}{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1 + 0 \times 0.8 \times 0.9 + 1 \times 0.2 \times 0.9}\\
&= 0.3382
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

R	P(R)
0	0.8
1	0.2

S	P(S)
0	0.9
1	0.1

Inference

$$
\begin{aligned}
&P(S=1|A=1)\\
&\propto P(S=1, A=1)\\
&\propto \sum_{B,R} P(A=1, B, R, S=1)\\
&\propto \sum_{B,R} P(A=1|R, S=1)P(B|R)P(R)P(S=1)\\
&\propto P(S=1)\sum_{R} P(A=1|R, S=1)P(R)\sum_{B} P(B|R)\\
&\propto P(S=1)\sum_{R} P(A=1|R, S=1)P(R)\\
&\propto 0.1(0.9\times 0.8 + 1\times 0.2)\\
&\propto 0.092
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

R	P(R)
0	0.8
1	0.2

S	P(S)
0	0.9
1	0.1

Inference

$$
\begin{aligned}
&P(S=0|A=1)\\
&\propto P(S=0, A=1)\\
&\propto \sum_{B,R} P(A=1, B, R, S=0)\\
&\propto \sum_{B,R} P(A=1|R, S=0)P(B|R)P(R)P(S=0)\\
&\propto P(S=0)\sum_{R} P(A=1|R, S=0)P(R)\sum_{B} P(B|R)\\
&\propto P(S=0)\sum_{R} P(A=1|R, S=0)P(R)\\
&\propto 0.9(0\times 0.8 + 1\times 0.2)\\
&\propto 0.18
\end{aligned}
$$

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

R	P(R)
0	0.8
1	0.2

S	P(S)
0	0.9
1	0.1

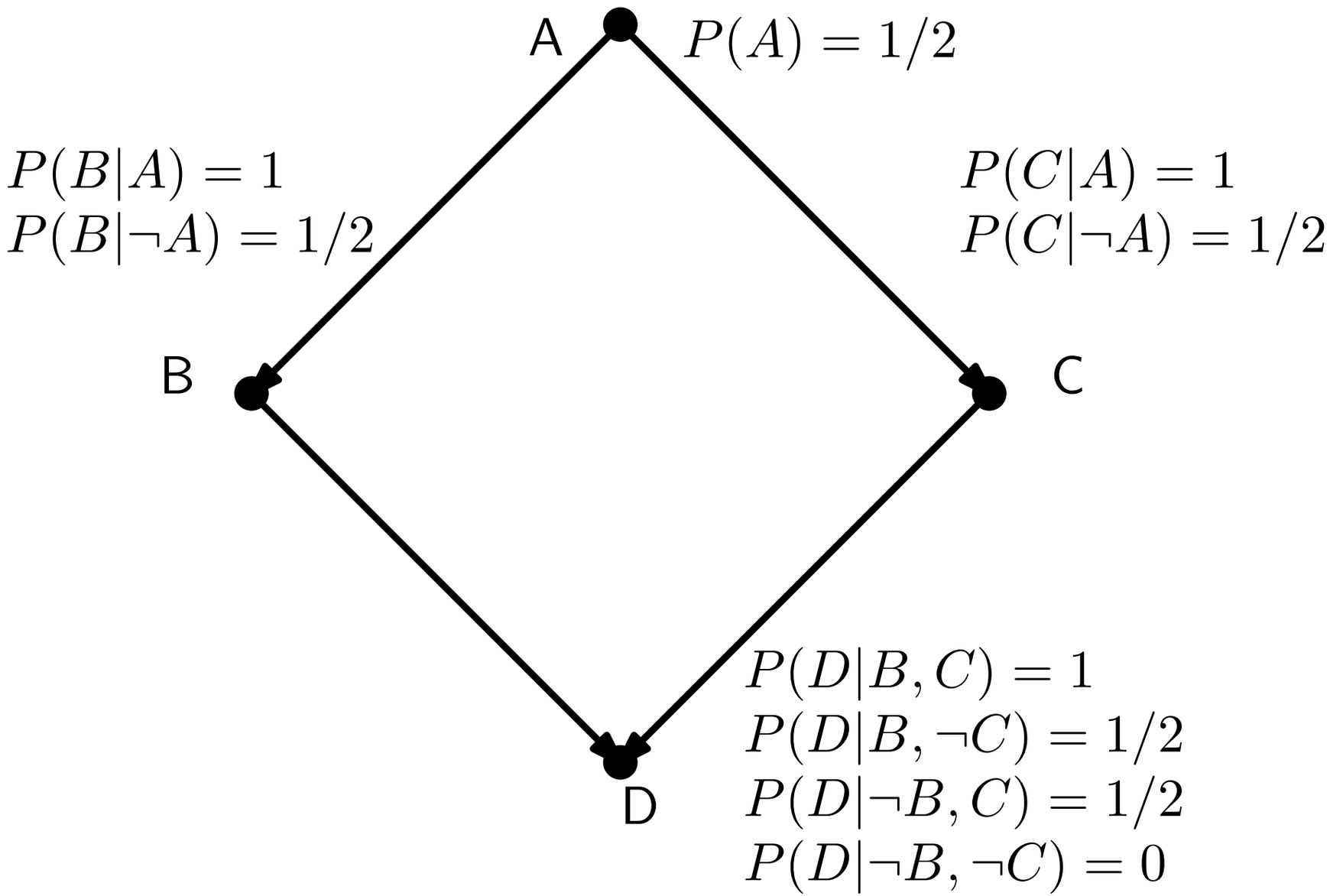
Inference

$$P(S = 1|A = 1) \propto 0.092$$
$$P(S = 0|A = 1) \propto 0.18$$

$$P(S = 1|A = 1) = \frac{0.092}{0.092 + 0.18} = 0.3382$$

$$P(S = 0|A = 1) = \frac{0.18}{0.092 + 0.18} = 0.6618$$

End


A
$P(A) = 1/2$
$P(B|A) = 1$
$P(B|\neg A) = 1/2$
$P(C|A) = 1$
$P(C|\neg A) = 1/2$
B
C
D
$P(D|B, C) = 1$
$P(D|B, \neg C) = 1/2$
$P(D|\neg B, C) = 1/2$
$P(D|\neg B, \neg C) = 0$